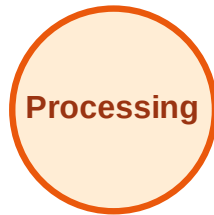


# BFS Traversal

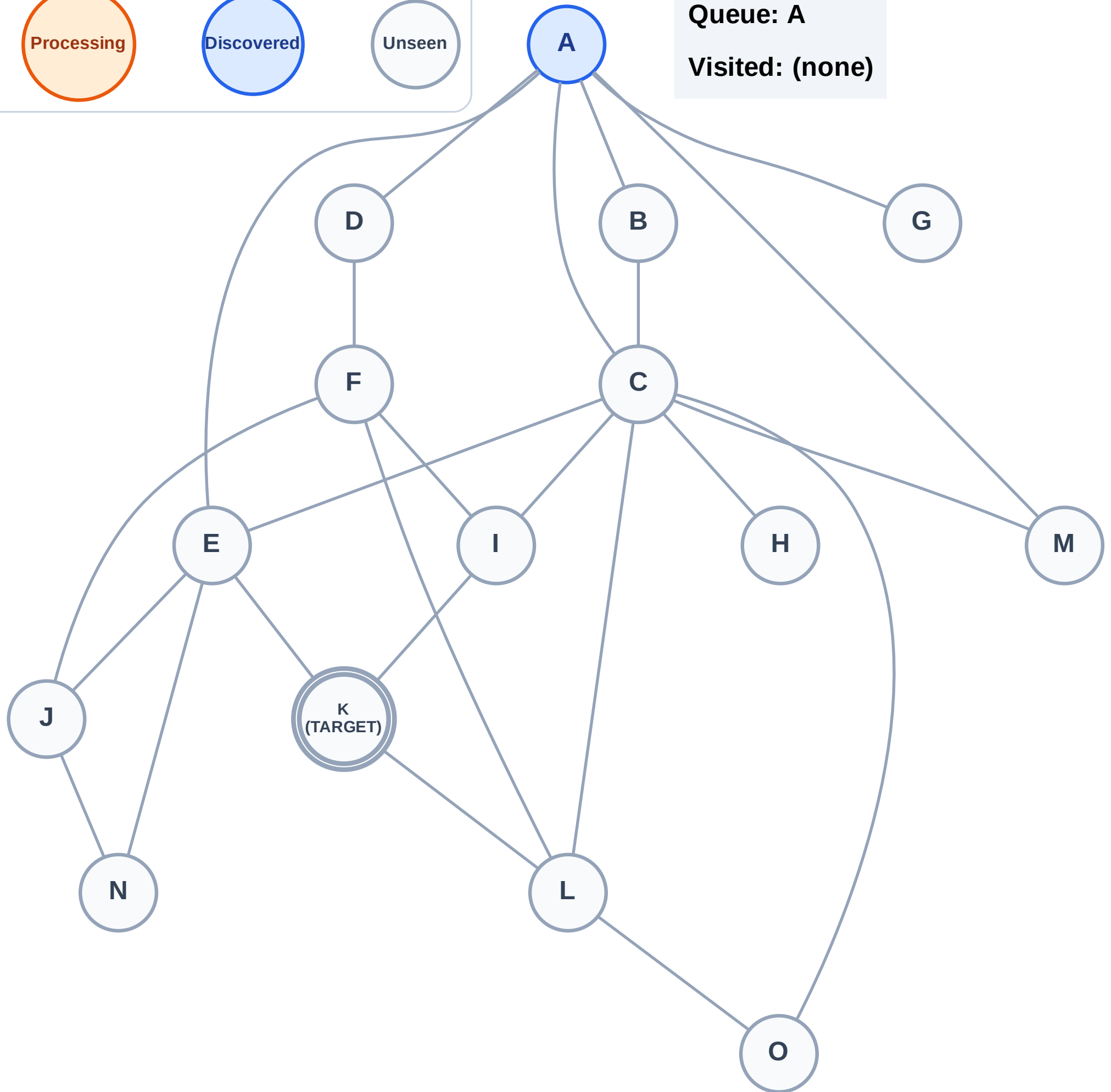
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

## Legend



Queue: A

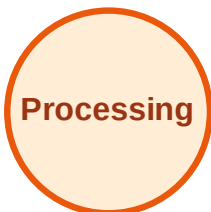
Visited: (none)



# BFS Traversal

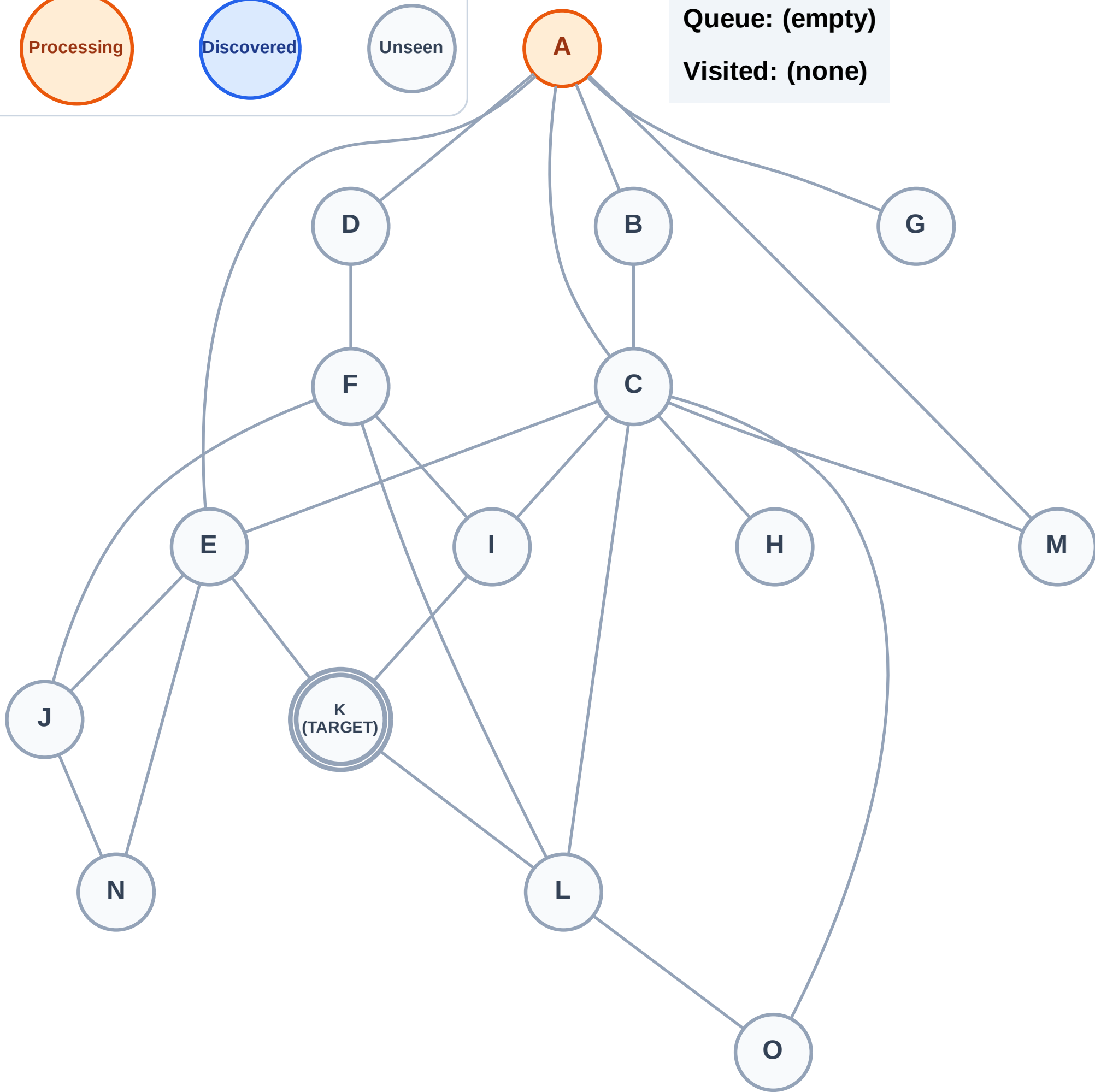
Step 2: Process node A

## Legend



Queue: (empty)

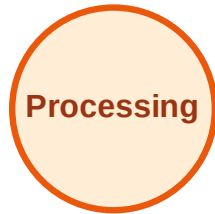
Visited: (none)



# BFS Traversal

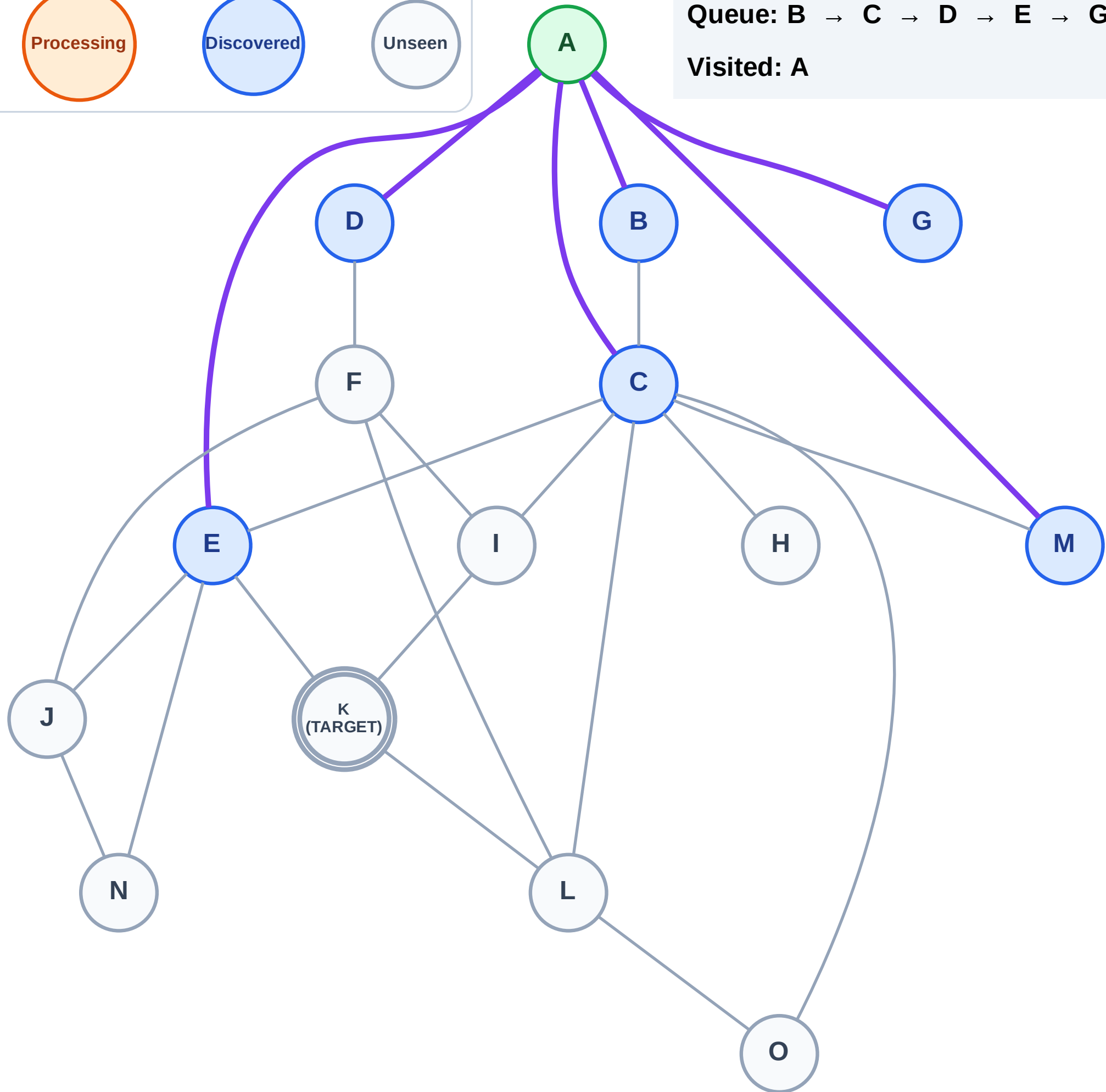
Step 3: Discover B, C, D, E, G, M from A

## Legend



Queue: B → C → D → E → G → M

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

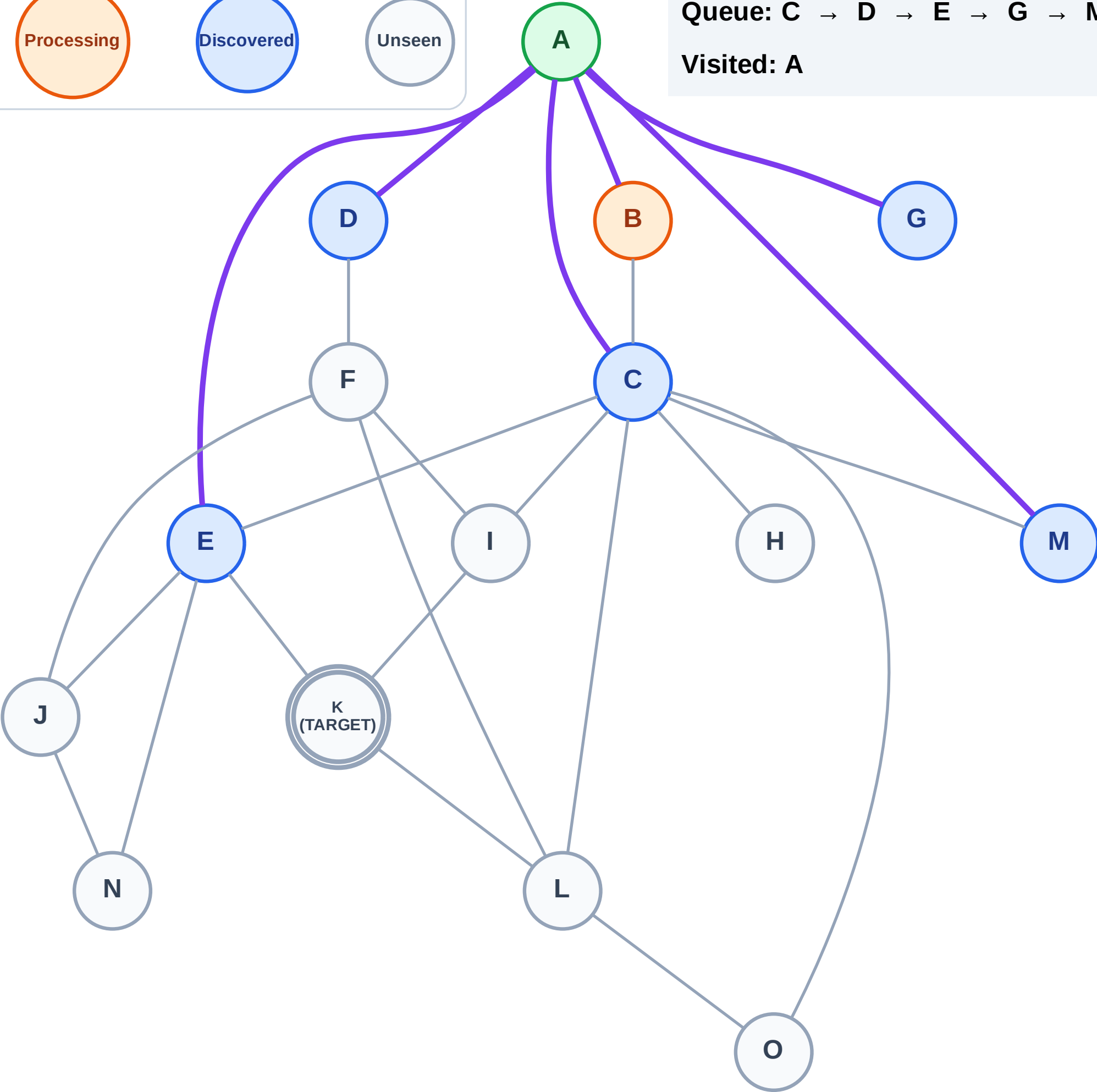
Processing

Discovered

Unseen

Queue: C → D → E → G → M

Visited: A



# BFS Traversal

Step 5: No new neighbors from B

Legend

Complete

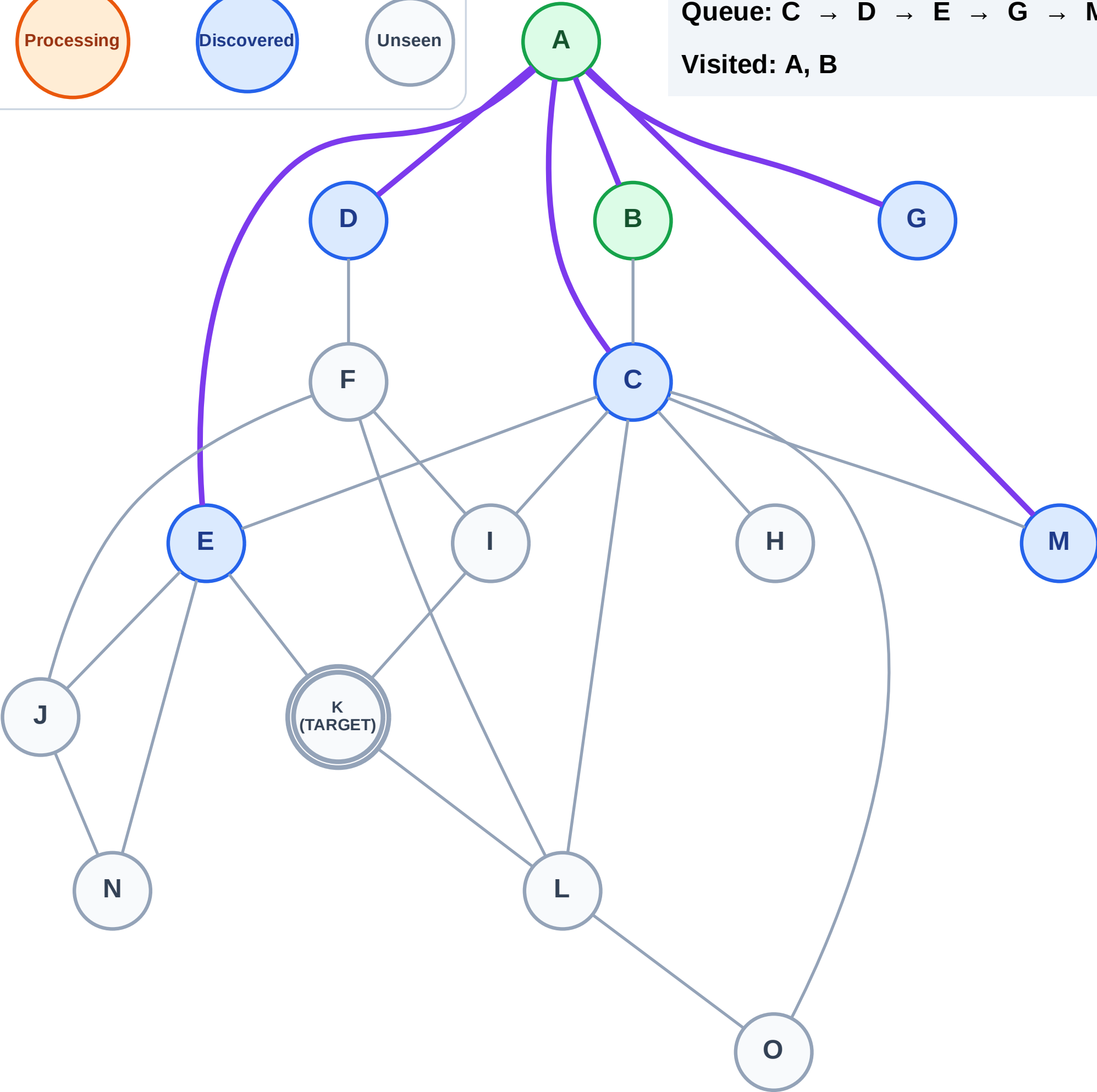
Processing

Discovered

Unseen

Queue: C → D → E → G → M

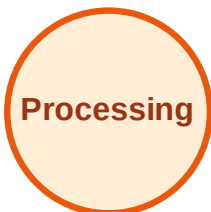
Visited: A, B



# BFS Traversal

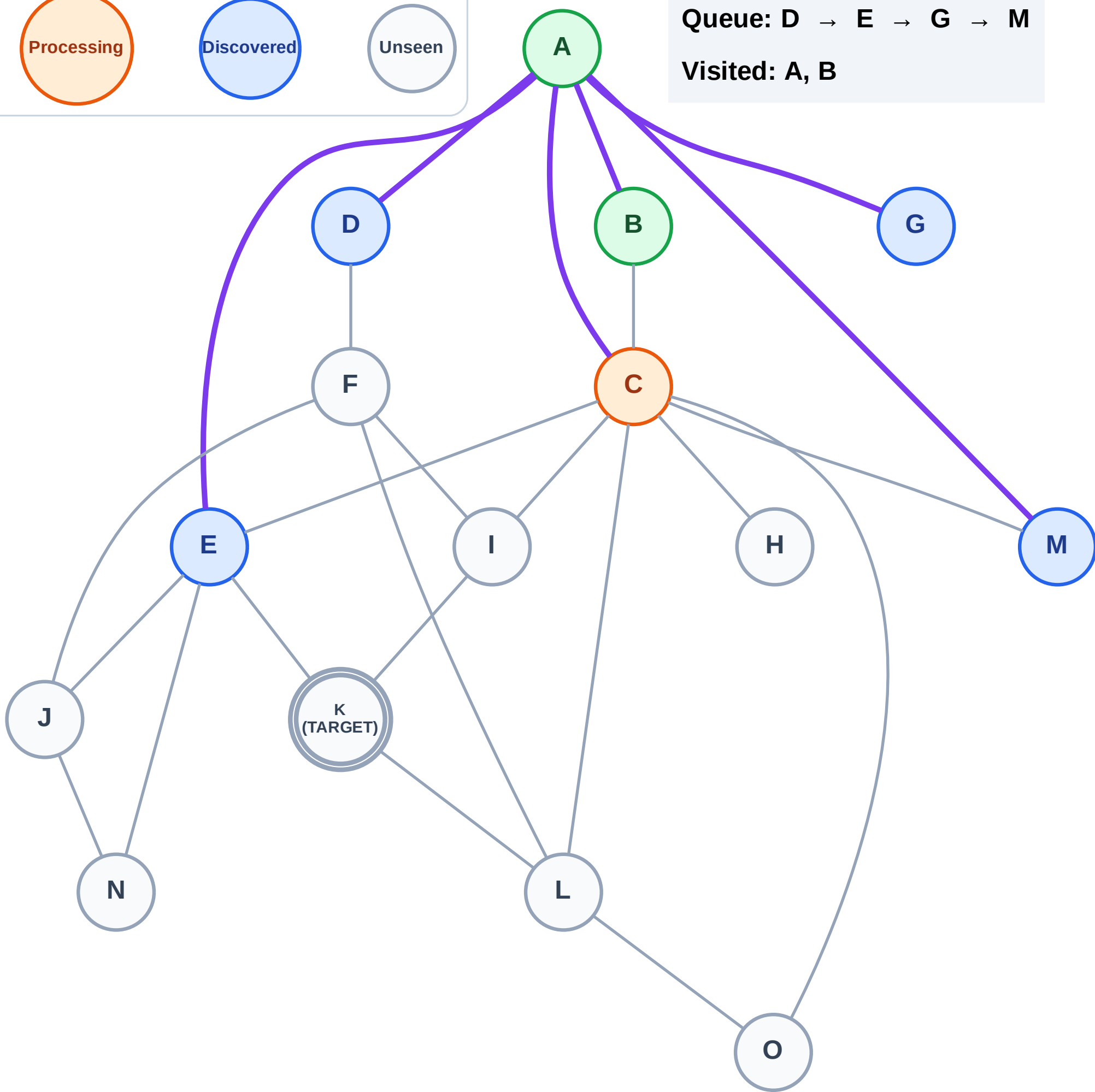
Step 6: Process node C

## Legend



Queue: D → E → G → M

Visited: A, B



# BFS Traversal

Step 7: Discover H, I, L, O from C

Legend

Complete

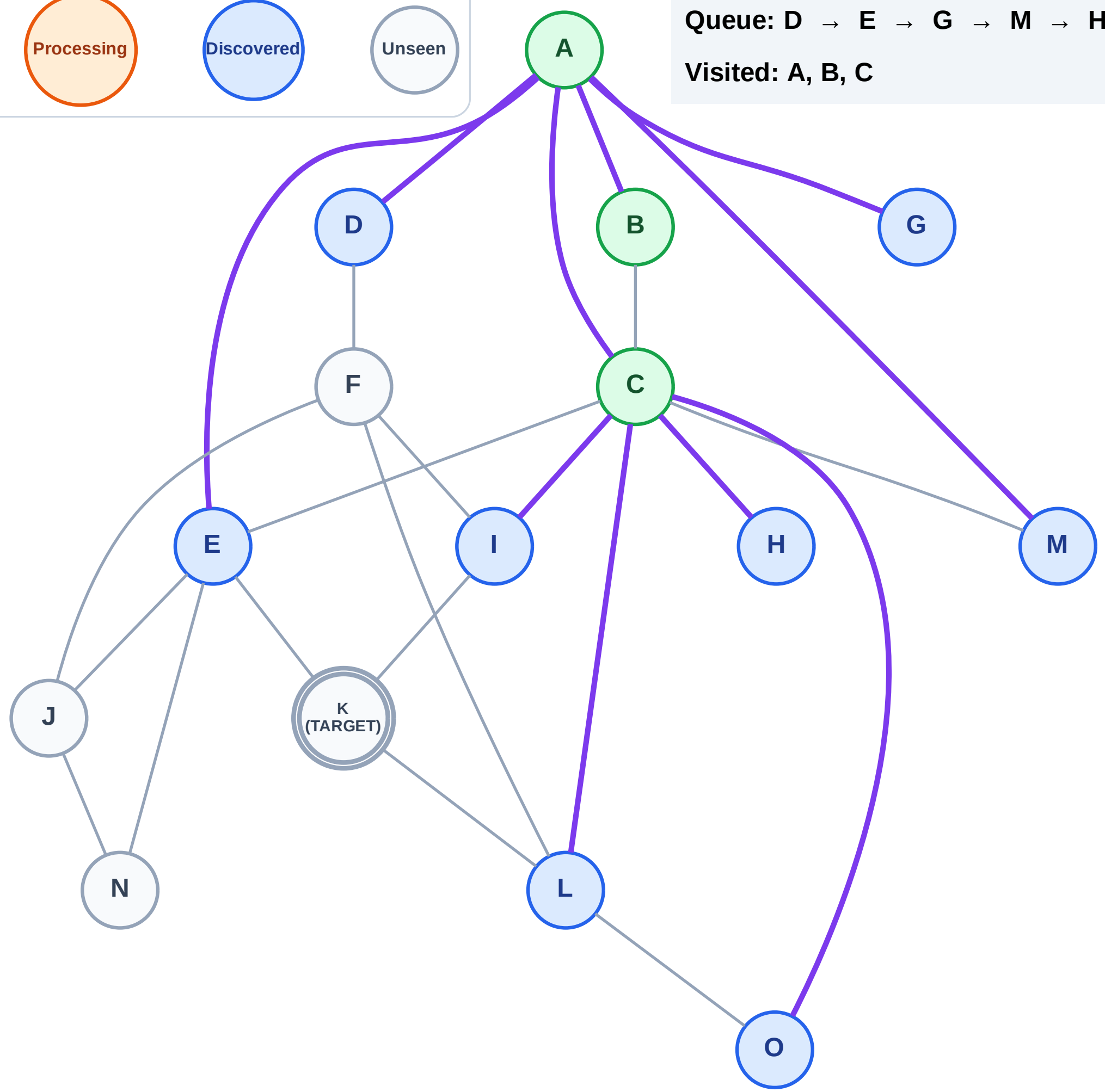
Processing

Discovered

Unseen

Queue: D → E → G → M → H → I → L → O

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

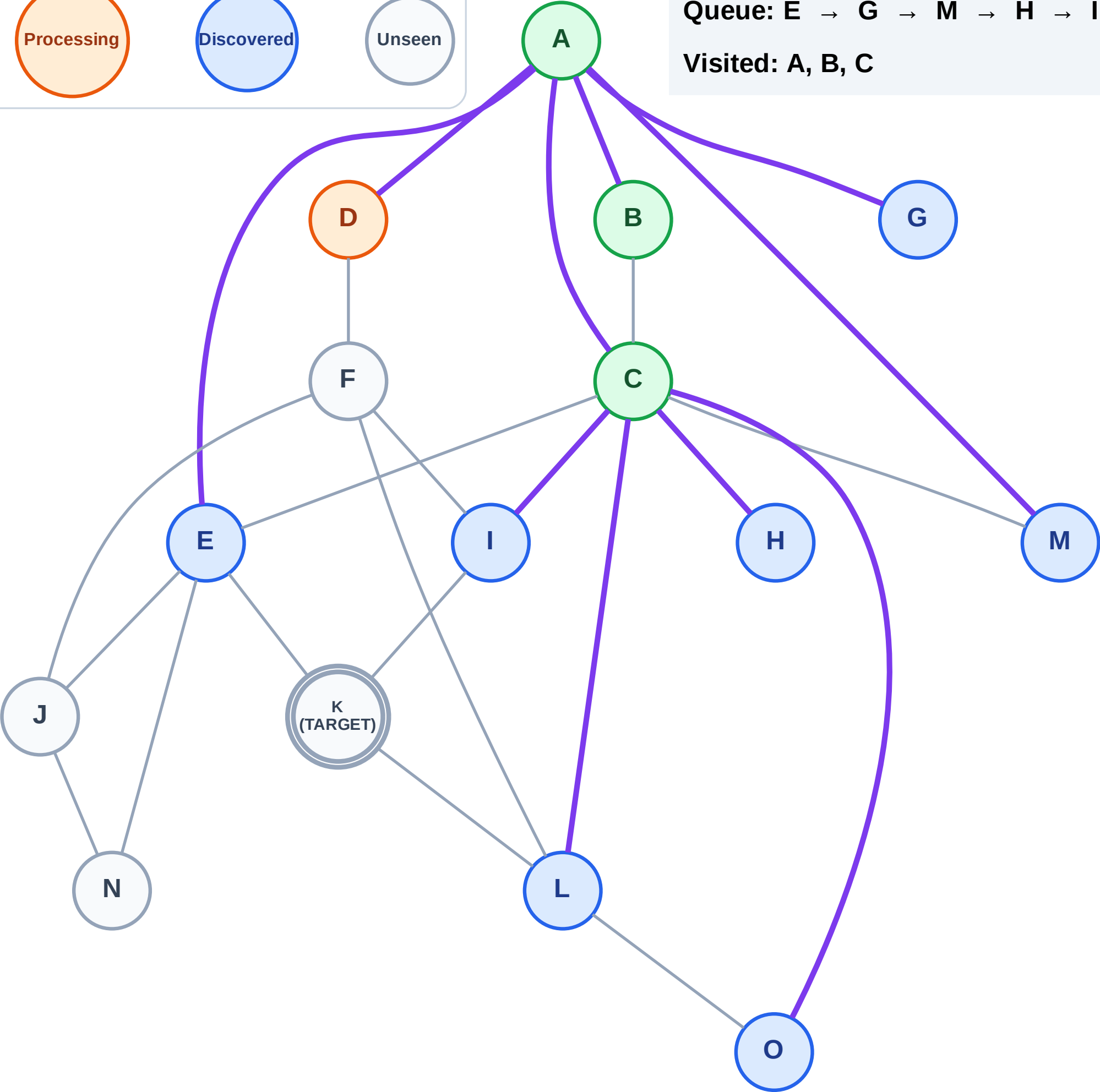
Processing

Discovered

Unseen

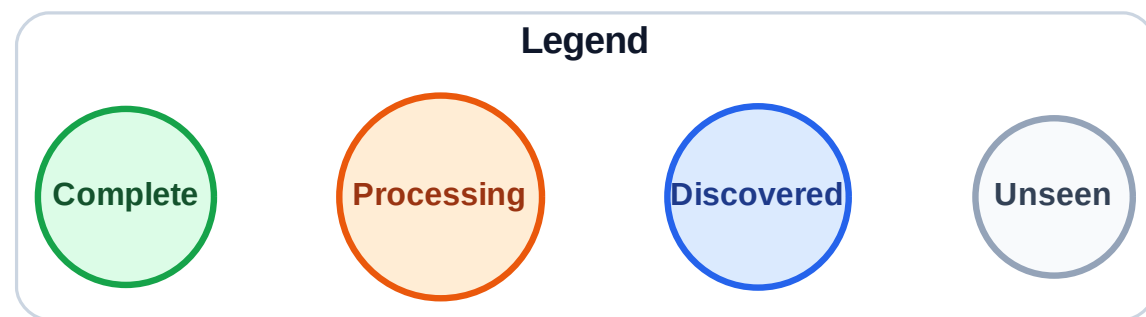
Queue: E → G → M → H → I → L → O

Visited: A, B, C



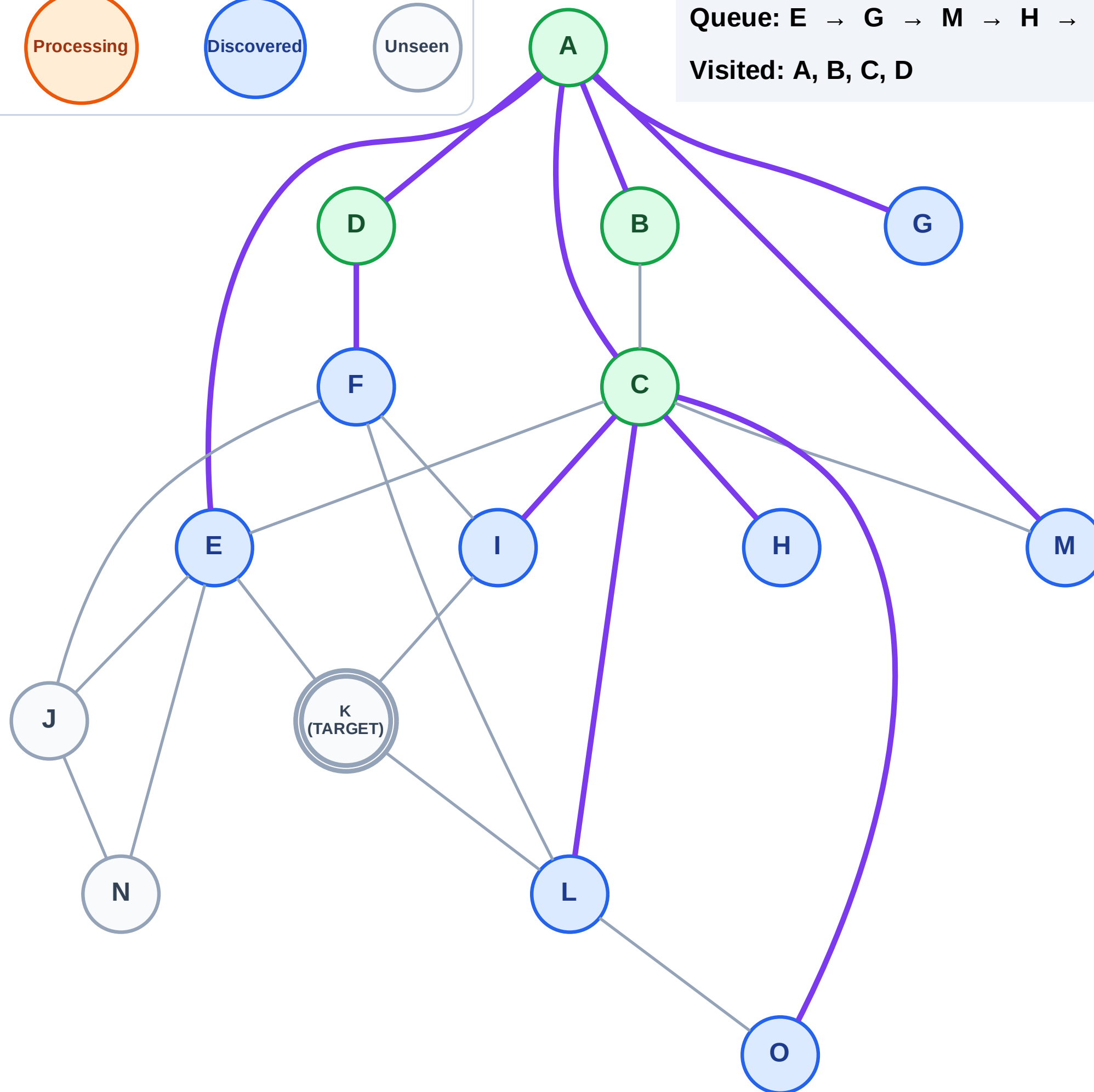
# BFS Traversal

Step 9: Discover F from D



Queue: E → G → M → H → I → L → O → F

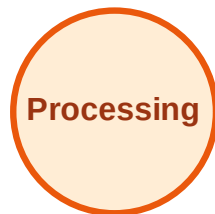
Visited: A, B, C, D



# BFS Traversal

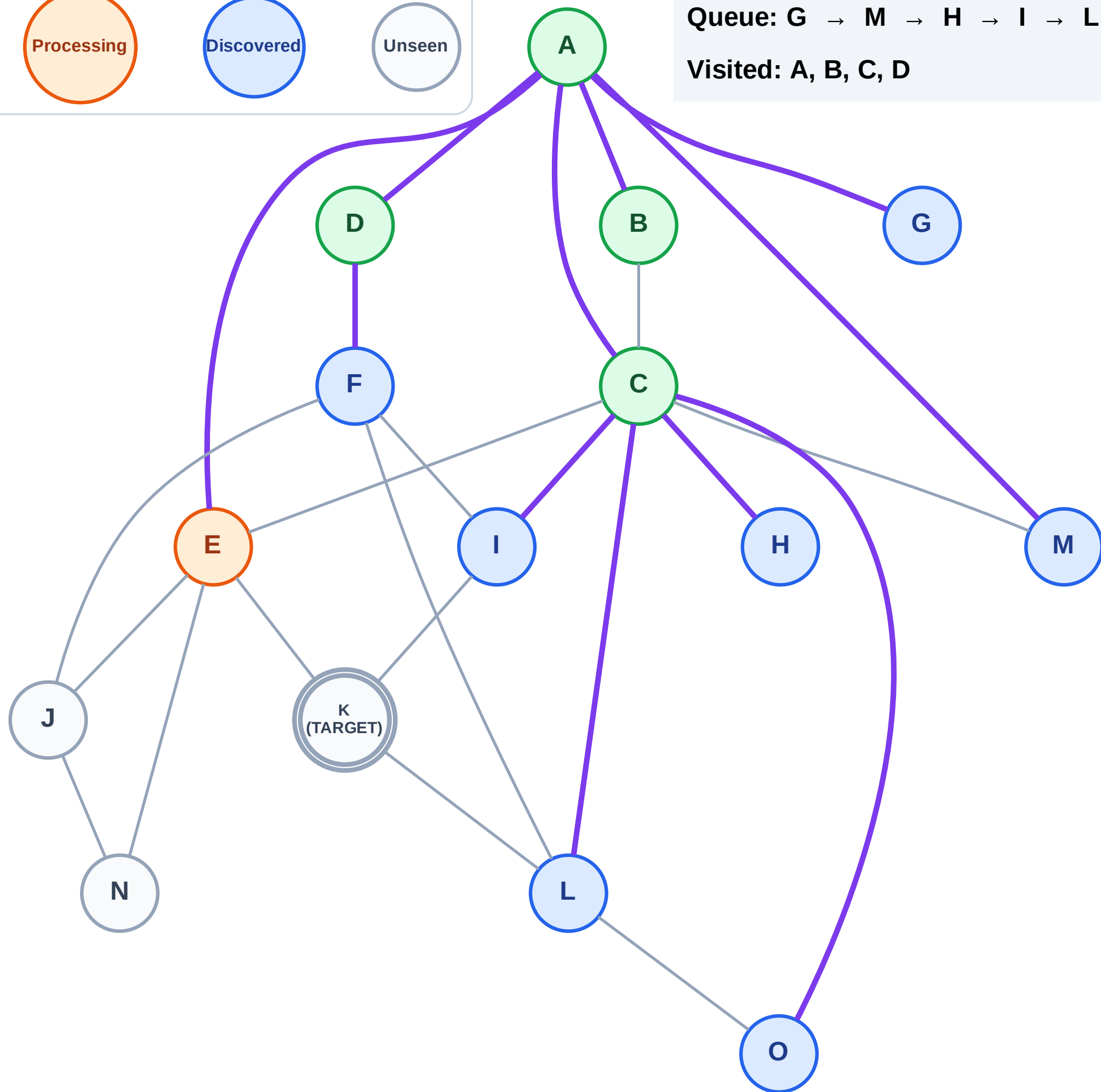
Step 10: Process node E

## Legend



Queue: G → M → H → I → L → O → F

Visited: A, B, C, D



# BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend

Complete

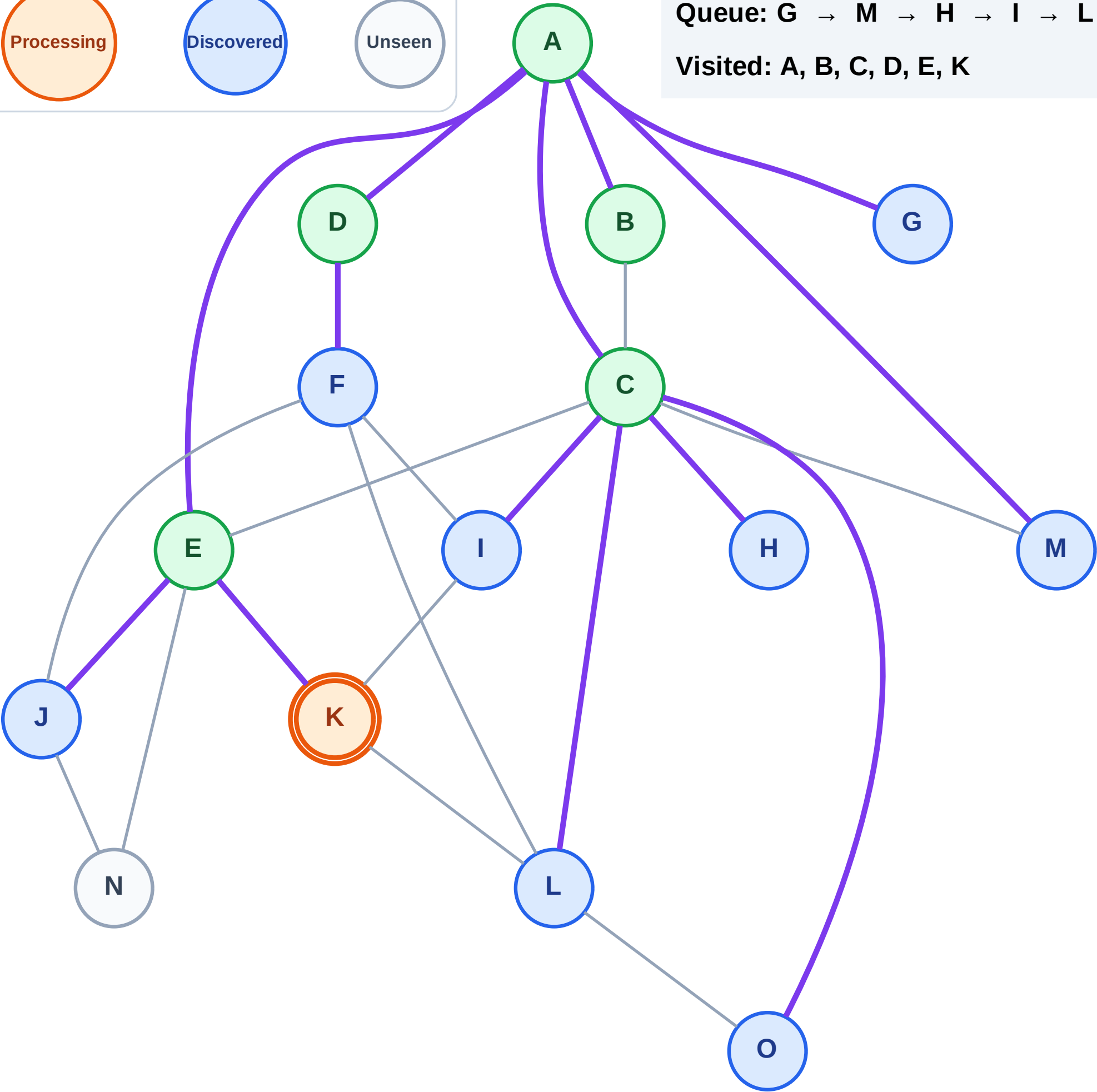
Processing

Discovered

Unseen

Queue: G → M → H → I → L → O → F → J

Visited: A, B, C, D, E, K



# DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

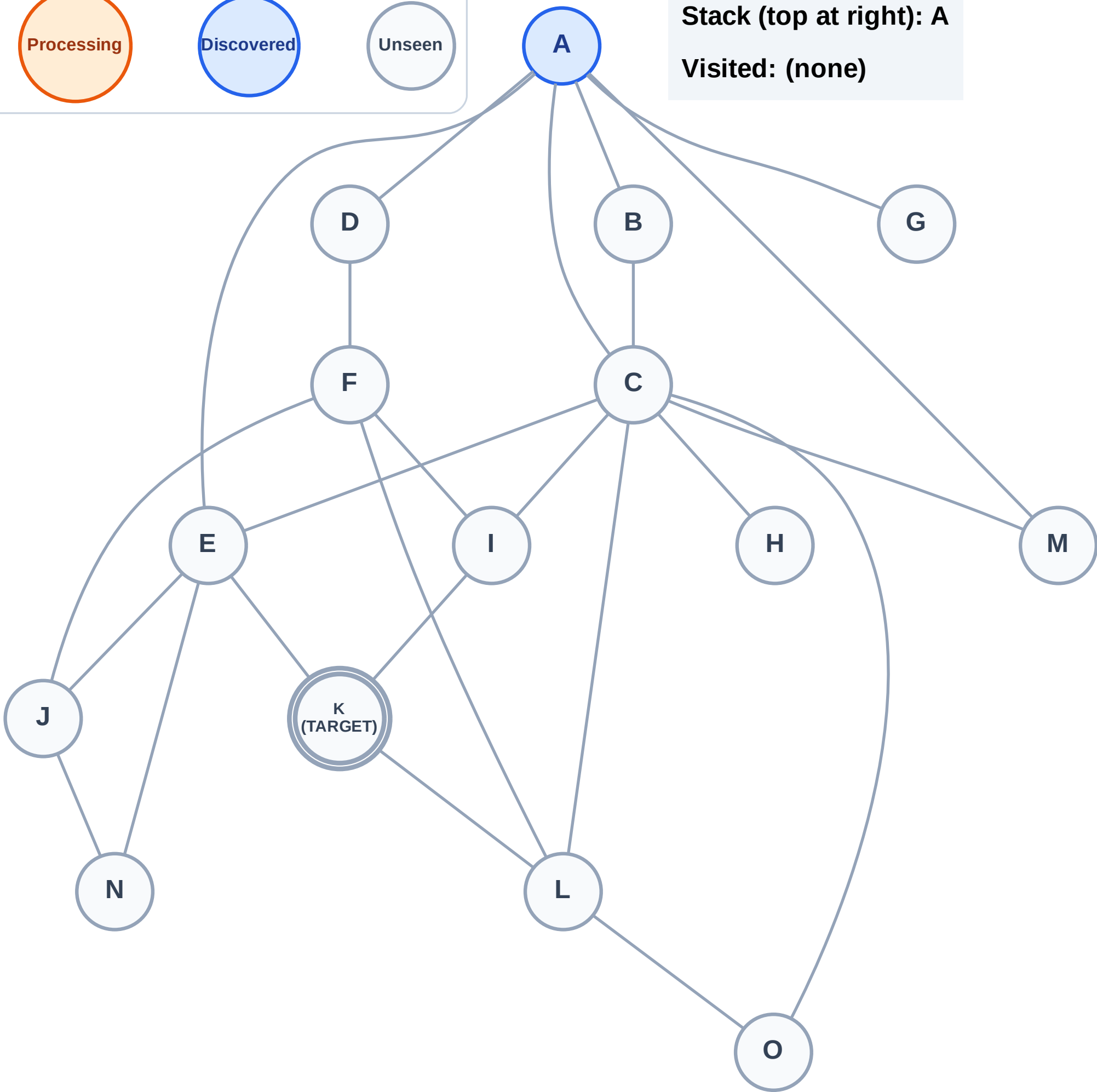
Complete

Processing

Discovered

Unseen

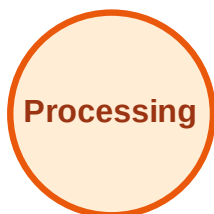
Stack (top at right): A  
Visited: (none)



# DFS Traversal

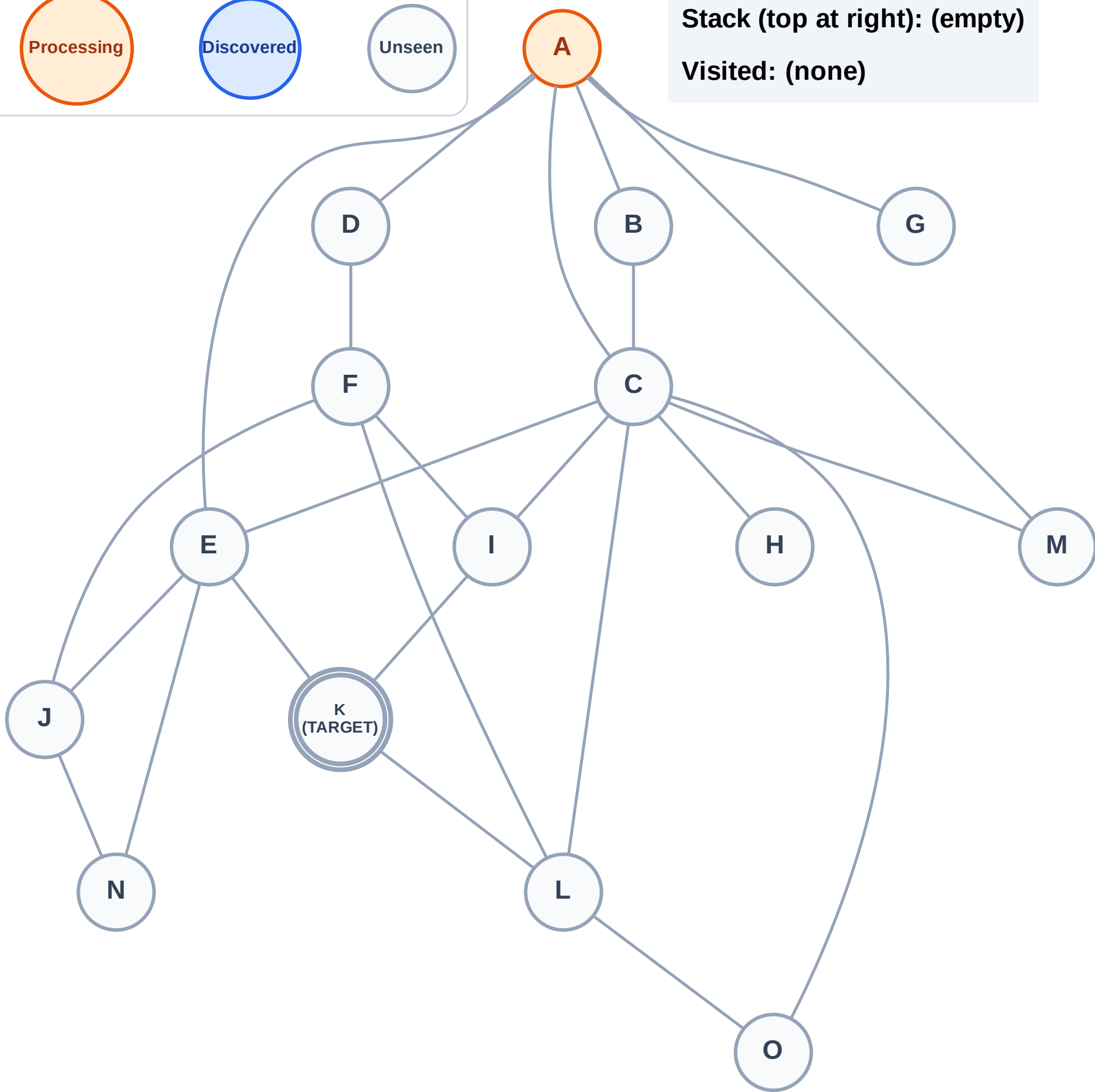
Step 2: Process node A

## Legend



Stack (top at right): (empty)

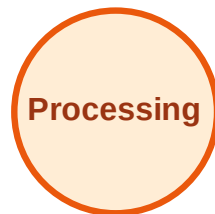
Visited: (none)



# DFS Traversal

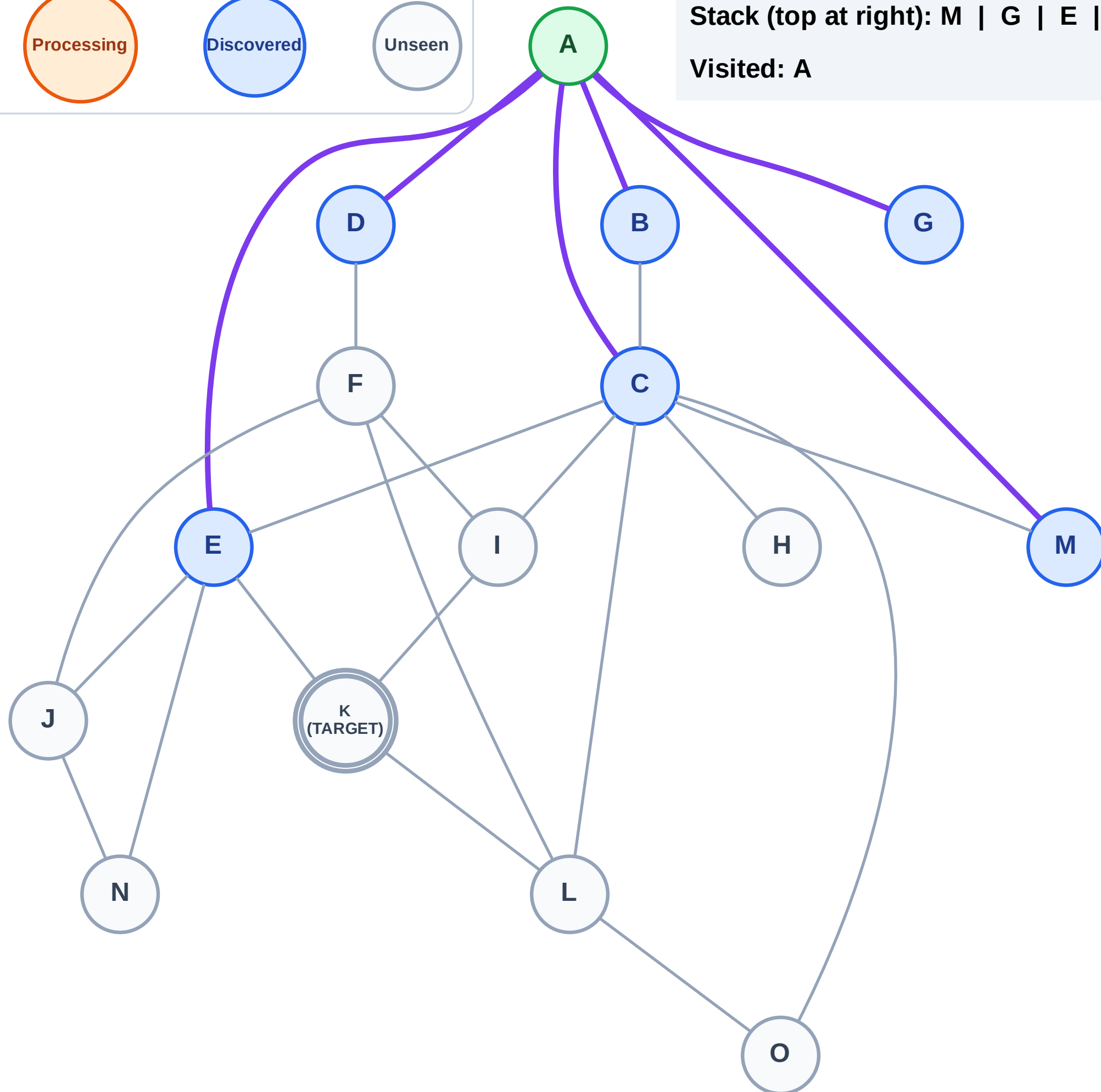
Step 3: Discover M, G, E, D, C, B from A

## Legend



Stack (top at right): M | G | E | D | C | B

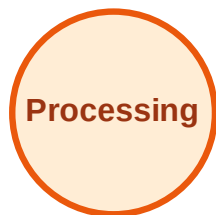
Visited: A



# DFS Traversal

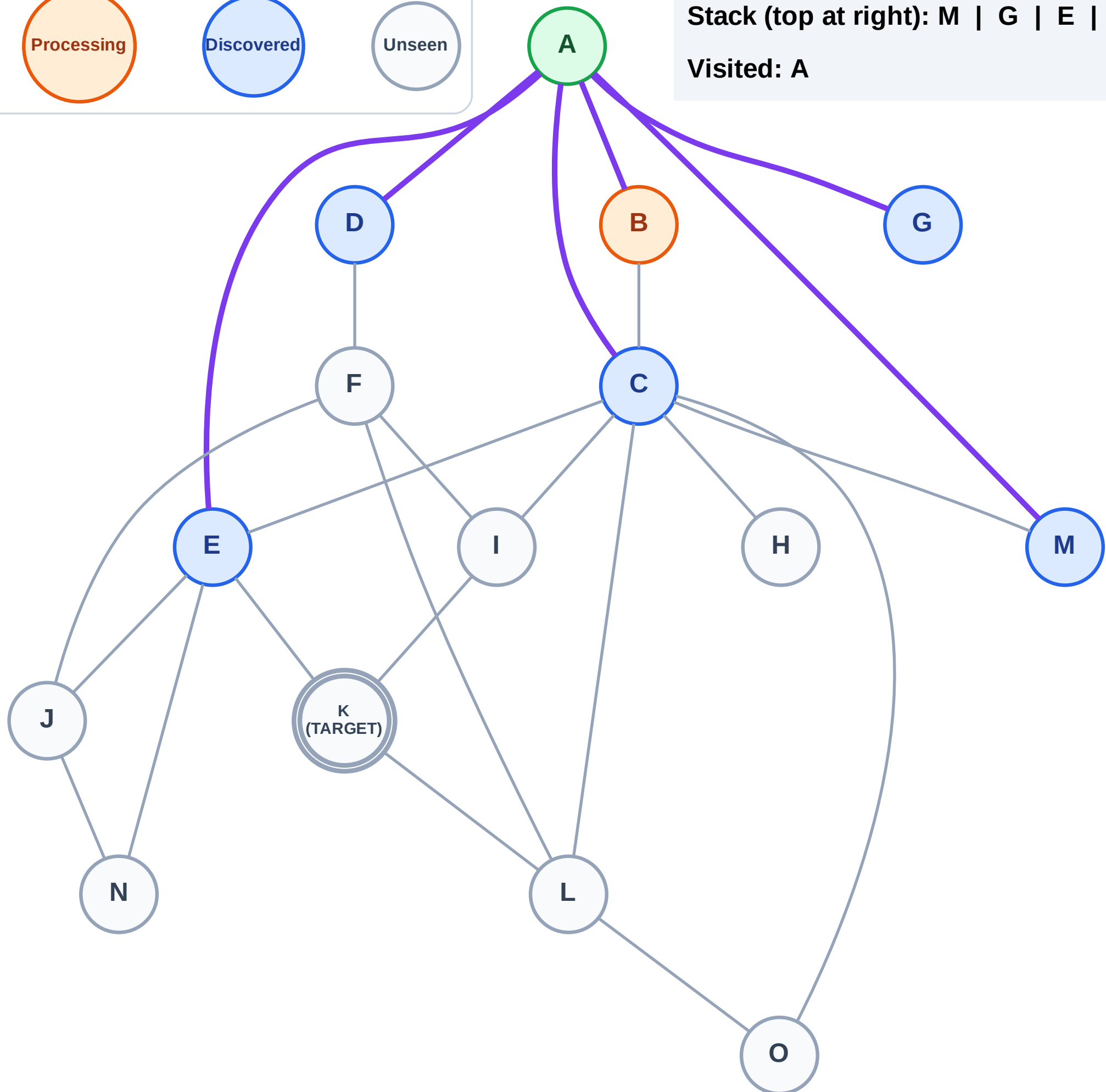
Step 4: Process node B

## Legend



Stack (top at right): M | G | E | D | C

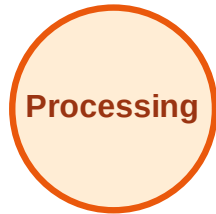
Visited: A



# DFS Traversal

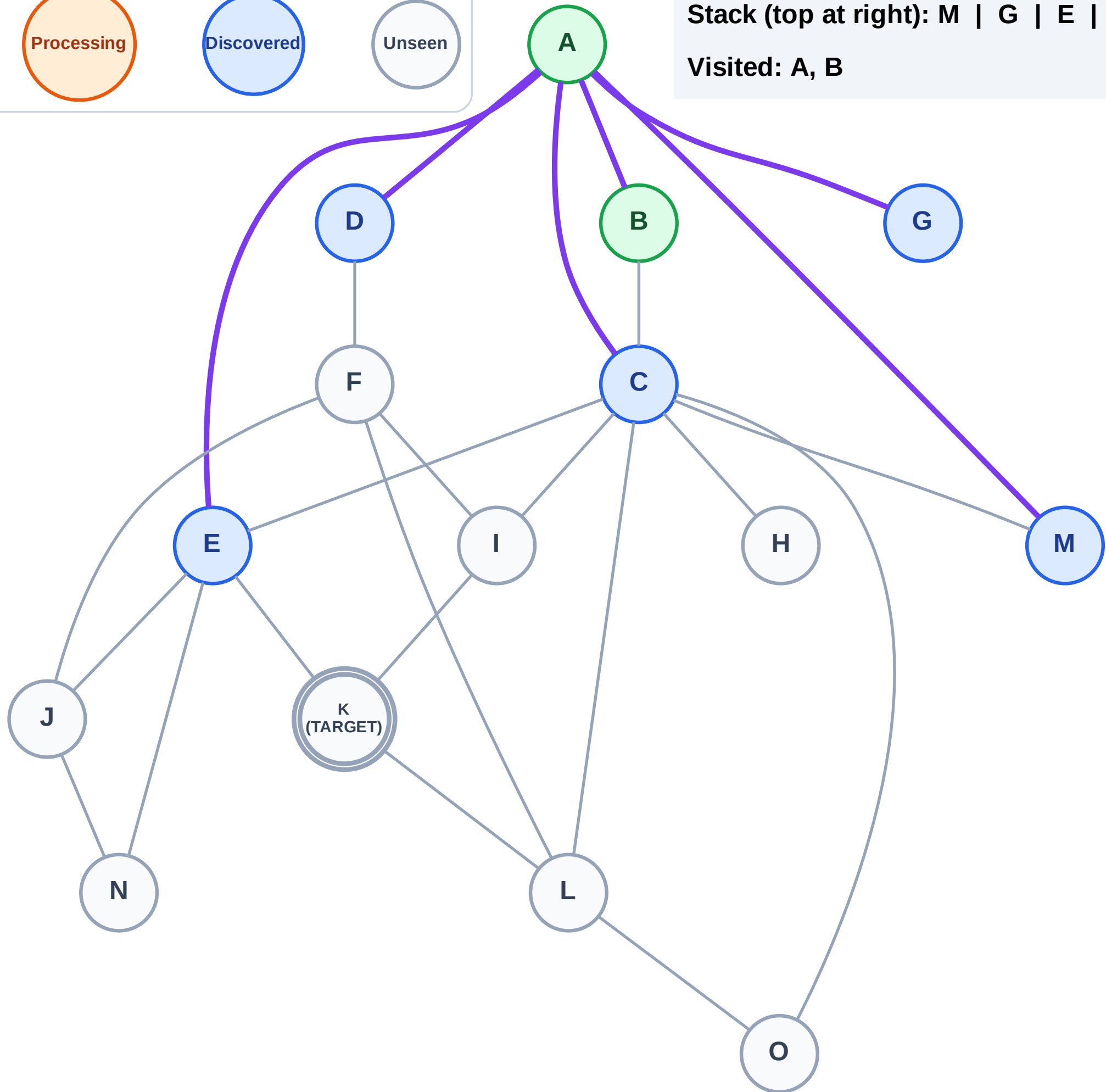
Step 5: No new neighbors from B

## Legend



Stack (top at right): M | G | E | D | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

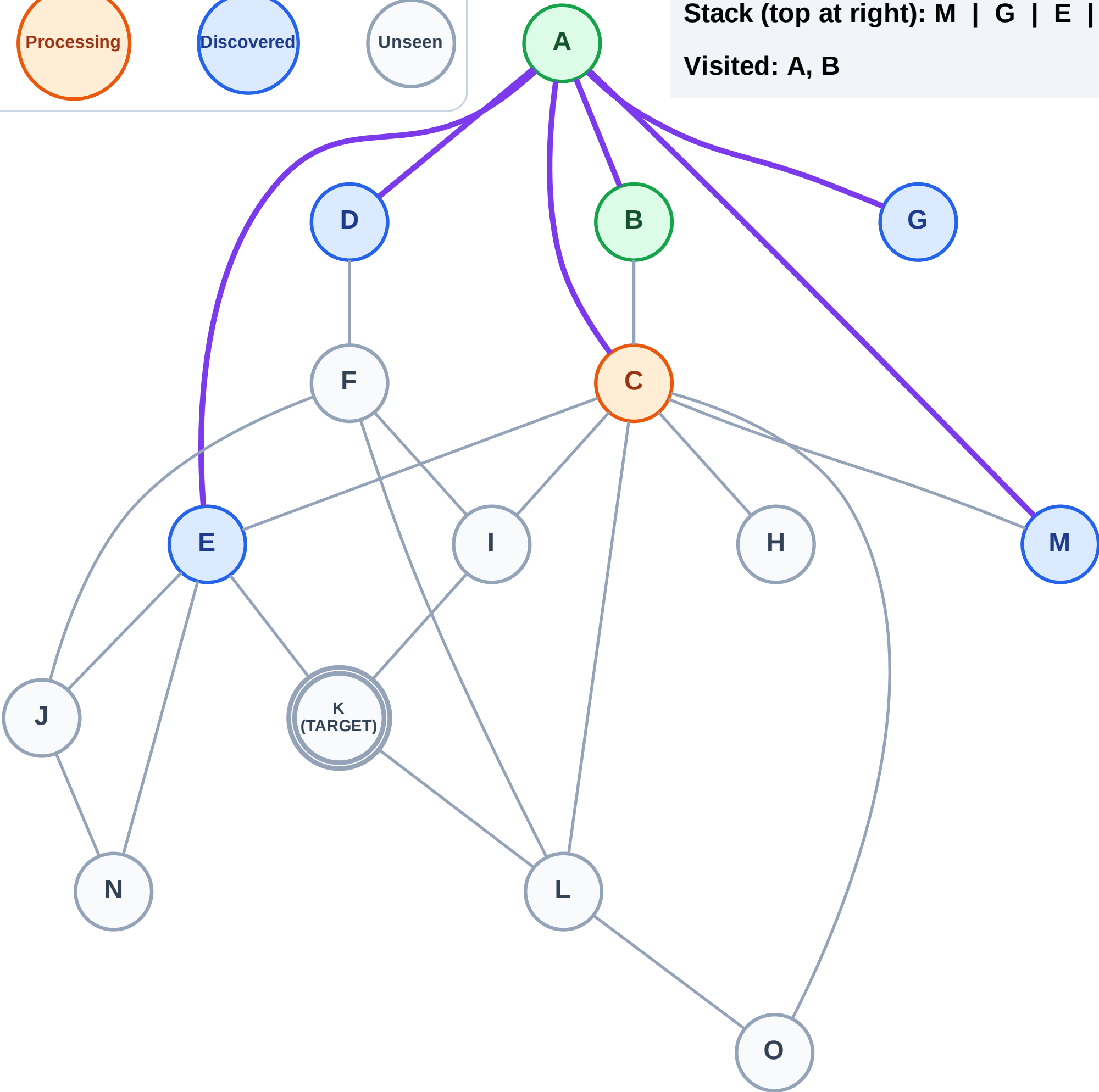
Processing

Discovered

Unseen

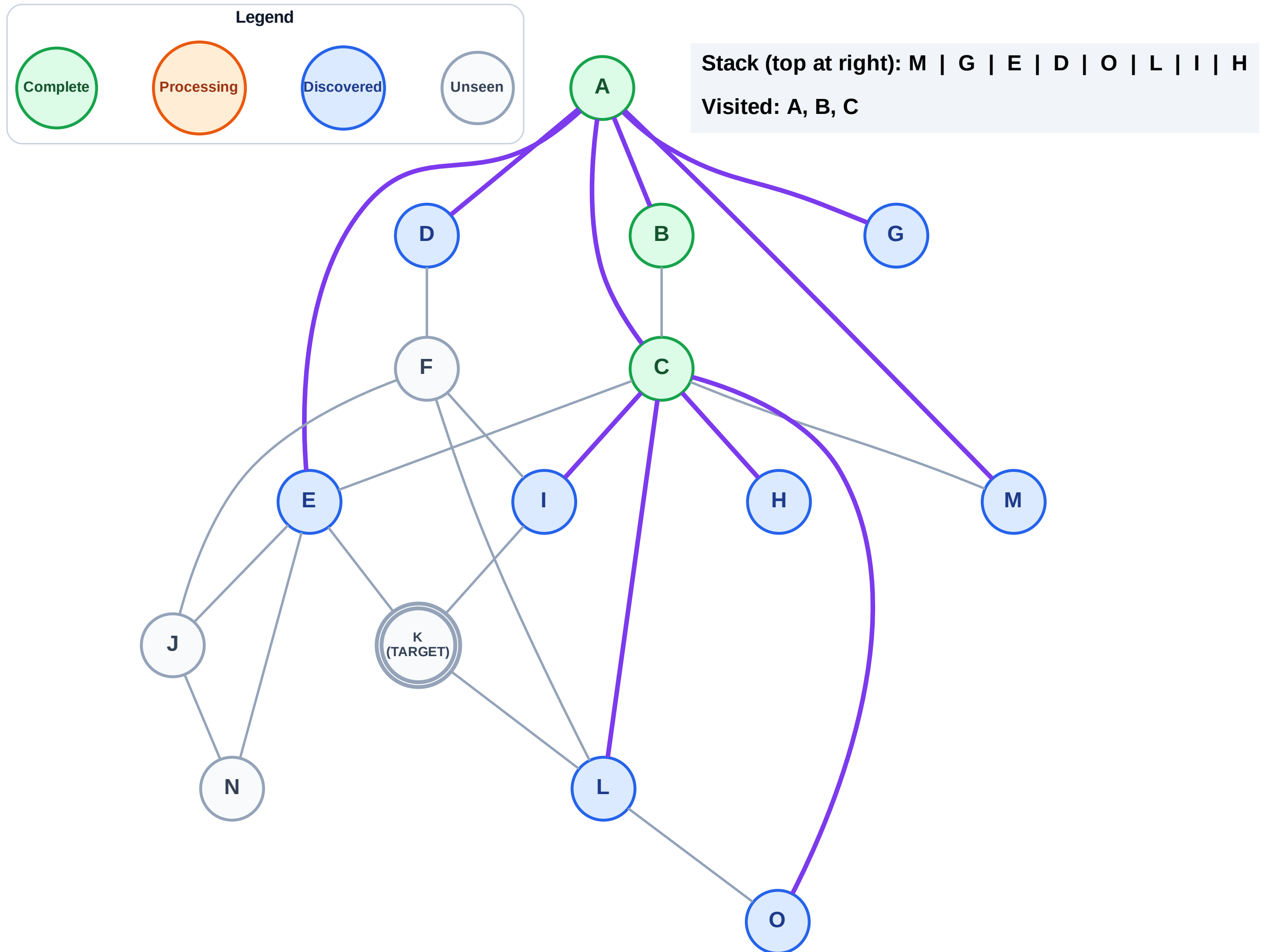
Stack (top at right): M | G | E | D

Visited: A, B



### Step 7: Discover O, L, I, H from C

### Step 7: Discover O, L, I, H from C



# DFS Traversal

Step 8: Process node H

Legend

Complete

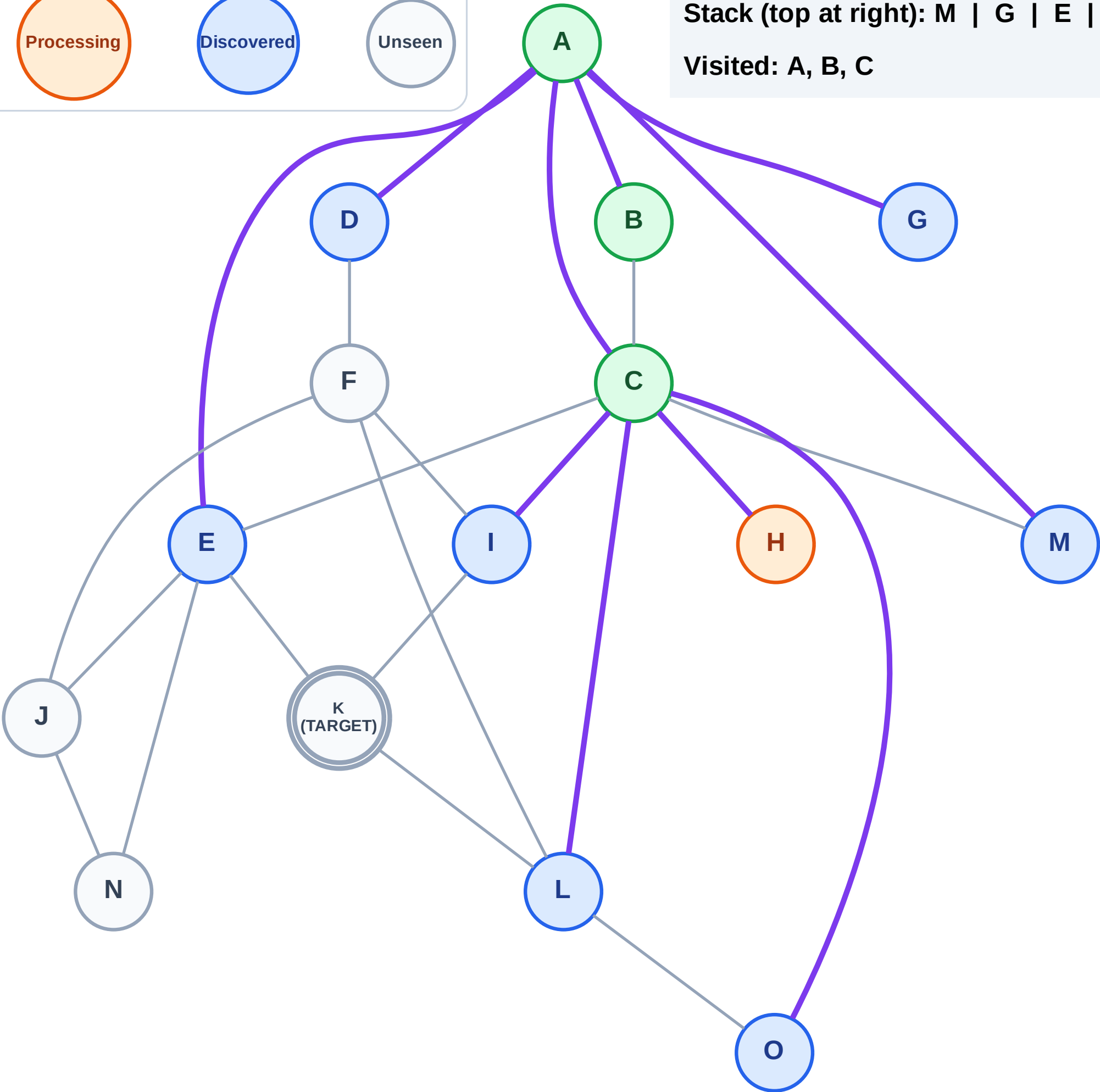
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | I

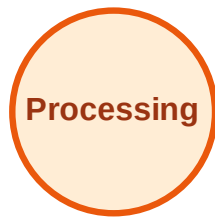
Visited: A, B, C



# DFS Traversal

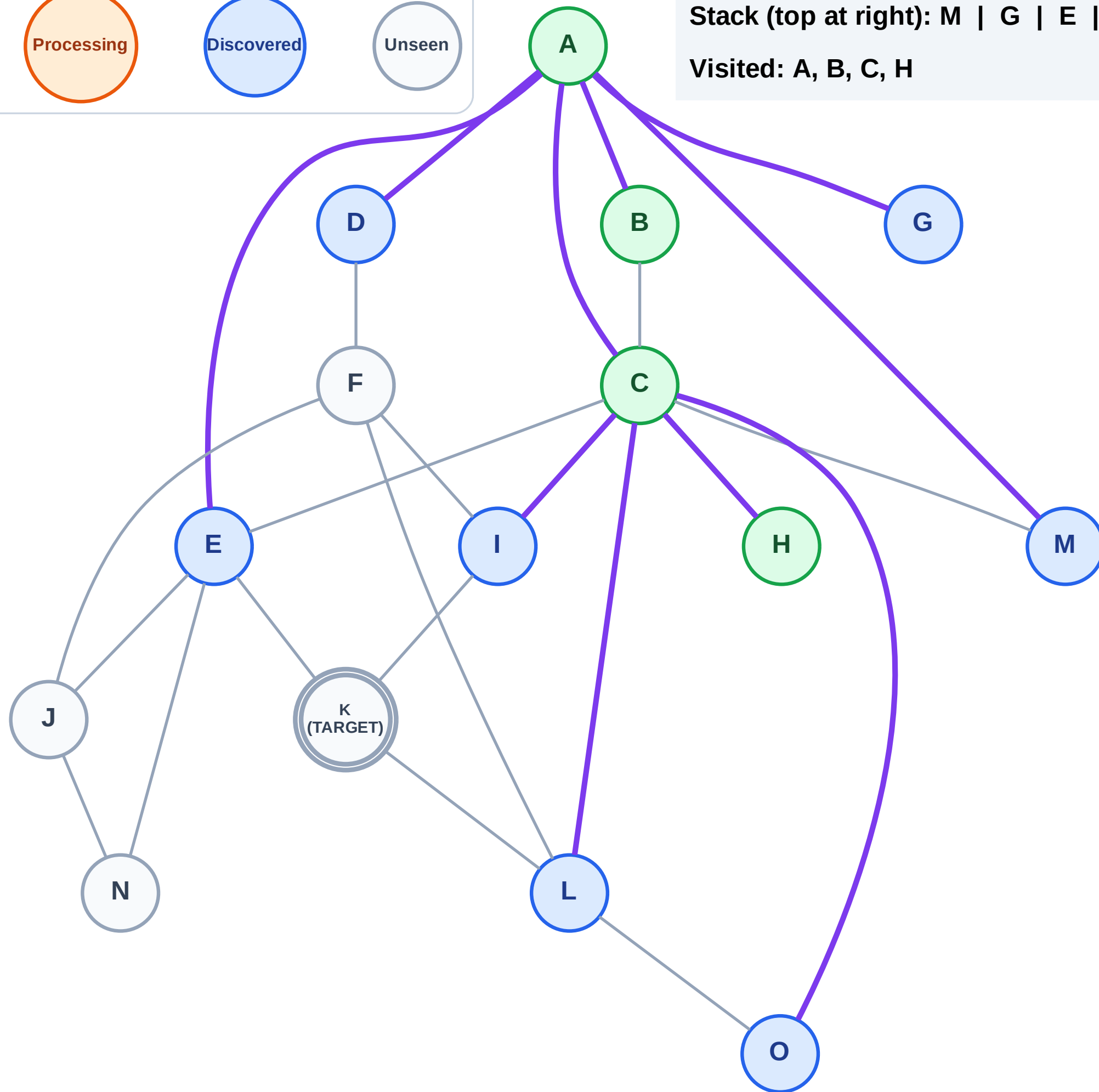
Step 9: No new neighbors from H

## Legend



Stack (top at right): M | G | E | D | O | L | I

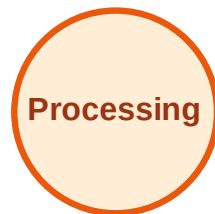
Visited: A, B, C, H



# DFS Traversal

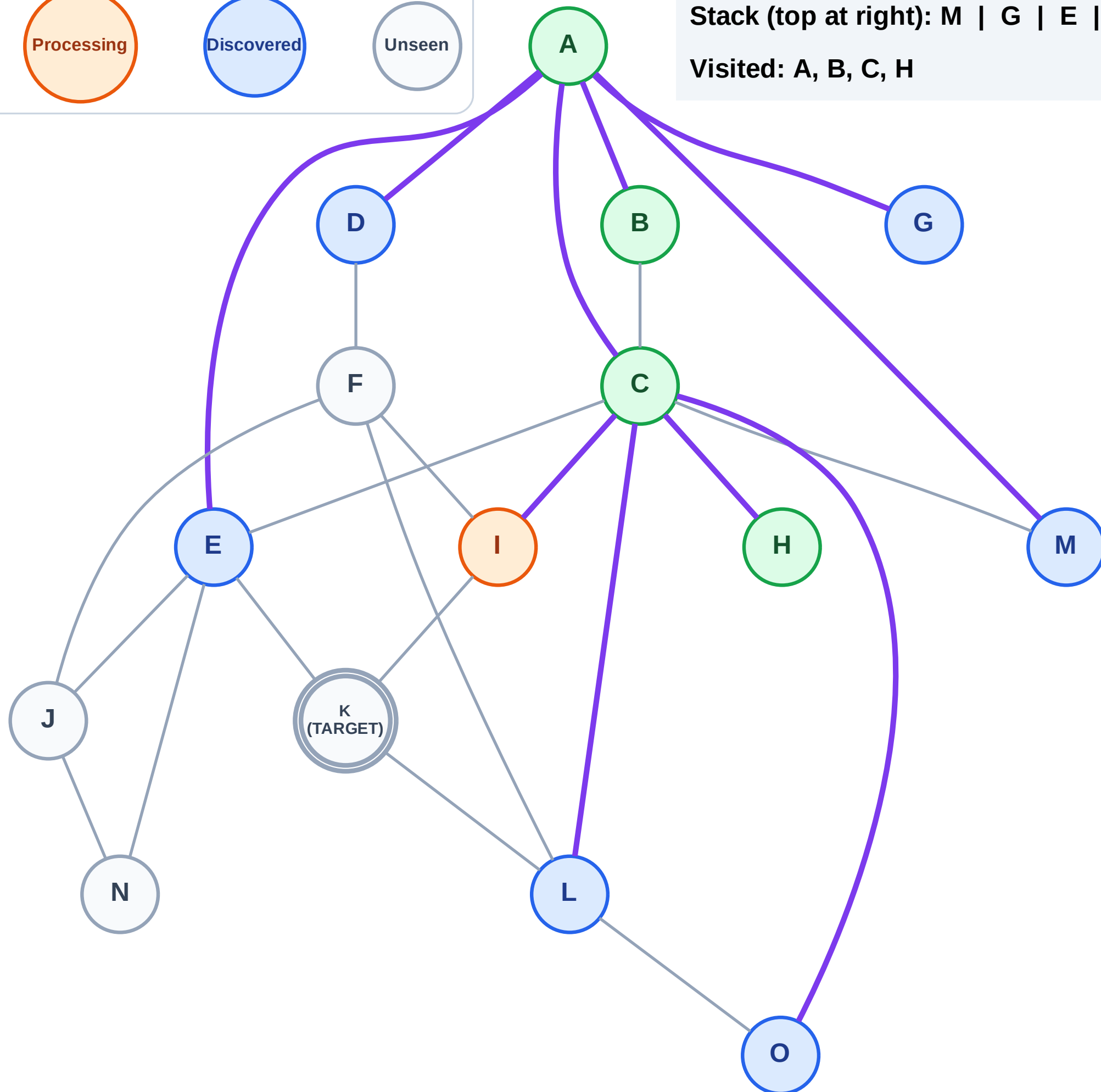
Step 10: Process node I

## Legend



Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, H



# DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → C → I → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, H, I, K

